

Yiran Liu

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SKILLS

- Proficient in data processing and task automation using **Python (numpy, pandas, requests)**, backed by solid understanding of algorithms and data structures.
- Skilled in object-oriented software development for complex systems using **Java** and **C++**.
- Experience with computer vision pipelines using **OpenCV** for image processing and **AprilTag** detection, as well as ML model deployment on edge devices (**jetson-inference**, **Ultralytics**).
- Experienced in collaborative software development lifecycle using **Git** and **CI/CD automation**.
- Working knowledge of common web frameworks (**Streamlit**, **FastAPI**, **Spring**, **Node.js**).
- Ability to manage relational databases using **SQL** queries and **SQLite** (Python).

EXPERIENCE

Software Engineering Lead & Team Captain | First Robotics Competition Sept. 2022 – May 2025

- Directed the team operation of two robotics teams as team captain, recognized with [Dean's List Award \(Vancouver, 2024\)](#) by the committee for strategic leadership and unwavering dedication.
- Led 30+ student engineers across both teams to achieve extraordinary innovations and engineering excellence, securing prestigious awards including the [Innovation in Control Award \(twice\)](#), [Autonomous Award](#) and [Engineering Inspiration Award](#).

Creator & Lead Maintainer | [MapleSim](#) (Open-Source Robotics Library) Nov. 2024 – Sept. 2025

- Developed MapleSim, a popular robotics simulation library with **500+** users world-wide.
- Led the collaborative development with **20+** international contributors, shipping **80+** feature updates and maintaining community-leading innovation standards.
- Authored **30-pages** of comprehensive [online documentation](#) with MKDocs, supporting **200+** student developers to successfully adopt the simulation technologies.
- Created **GitHub Actions** workflows to automate development processes including branch merging, code formatting, distribution publishing, and **Javadoc** generation.
- Presented community-leading innovations at the FIRST global robotics conference upon committee invitation, inspiring **200+** youth developers.

PROJECTS

["MapleSim"](#) - Open-Source Robotics Library | [WPILib](#) (Java) Nov. 2024 – Sept. 2025

A popular robotics code library delivering robotics simulation technology using a rigid-body physics engine. Used by 500+ teams worldwide in the FIRST Robotics Competition, earning 70+ stars on GitHub.

["Phantom-Vision"](#) - Robotics Vision System | [NVIDIA Jetson](#) (Python) Feb. 2024 – Sept. 2024

An open-source, budget-friendly computer vision solution for robotics teams, featuring robot navigation using **AprilTags** and AI object detection using **Jetson-Inference**. Deployed on multiple competition robots with a dataset labelling toolkit based on **RoboFlow** for model training, and web-based telemetry dashboard using **Python**. Reduced vision system costs by **70%**, saving **\$1000+** for 6 beginner teams.

["IntelliScope"](#) - FRC Scouting Software | [Streamlit](#) + [SQLite](#) (Python) In Progress

An interactive web dashboard for scouting during the FIRST Robotics Competition. Monitors team performance, processes match data, and visualizes key statistics for strategic analysis.